$= 2 \times 10^{-11}$ cm^3 molecule^{-1} s^{-1} [8,9]). This should be compared with the lifetimes before reaction with OH and NO_3, which are both ca. 30 h ([OH] $\simeq 10^6$ molecules cm^{-3}, k(OH + DMS) = 8×10^{-12} cm^3 molecule^{-1} s^{-1};[43] [NO_3] $\simeq 10^7$ molecules cm^{-3}, k(NO_3 + DMS) = 9×10^{-13} cm^3 molecule^{-1} s^{-1} [44]). This suggests that IO may be a potentially important oxidizing radical in the marine boundary layer. The impact of halogen oxides on DMS oxidation has been discussed recently by Barnes et al.[45]

Although there as been some progress in recent years in measuring rate constants for iodine reactions of potential atmospheric importance, there are still many uncertainties, and it is possible that the above conclusions may be changed dramatically as key parameters are determined. It appears, however, that sink routes involving reactions of IO are likely to be important. In this respect, the reaction of IO with HO_2 needs to be studied in addition to confirmation of the single measurement of the IO + NO_2 reaction.[11] Detection of the products HOI, $IONO_2$, and I_2O_2 is also of great importance, together with measurement of their thermal decomposition rates and characterization of their UV-visible absorption spectra to assist in determining their atmospheric lifetimes before photolysis.

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Krypton Chloride and Krypton Fluoride Thermochemistry and Formation and Relaxation Kinetics in Ar, N_2, and CF_4 Buffer Gas

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Krypton chloride and krypton fluoride molecules were formed by the Kr($^3P_1$) sensitization reaction with Cl_2 or F_2 in variable pressures (up to 5 atm) of Ar, N_2, and CF_4 buffer gases at 300 and 230 K. Less extensive experiments also were done in He and Ne. The high-pressure (equilibrium) KrF(C)/KrF(B) and KrCl(C)/KrCl(B) ratios were observed in order to assign the energy separation of the C and B states. Although not totally conclusive, the present data support ΔE_{BC} ($=E_C - E_B$) values of -120 ± 50 and 0 ± 50 cm^{-1} for KrF* and KrCl*, respectively. The KrF(B,C) + 2Ar rate constant for ArKrF* formation was assigned as $(8 \pm 3) \times 10^{-33}$ cm^6 molecule^{-2} s^{-1} at 300 K; a slightly larger value was estimated for formation of ArKrCl*. Three-body formation of ArKrF* and ArKrCl* in Ar buffer was enhanced by reduction of the temperature. The increase in KrF(D) and KrCl(D) concentration, relative to KrF(B,C) and KrCl(B,C), with increased Ar pressure is attributed to the larger three-body quenching rates for the KrF(B,C) and KrCl(B,C) states. The C → B collisional transfer and the vibrational relaxation of KrF(B,C) and KrCl(B,C) also are discussed.

I. Introduction

The Kr($^3P_1$) sensitization technique with Kr/X_2 (X = F, Cl) mixtures with added buffer gas provides a steady-state method for studying the relaxation and quenching of KrF(B,C) and KrCl(B,C) molecules from the low-pressure, collision-free limit up to ~5 atm pressure.[1-5] Early work[1,4] at low pressure established that Xe($^3P_1$) and Xe($^3P_2$) or Kr($^3P_1$) and Kr($^3P_2$) reacted in similar ways with halogen-containing reagents with respect to product branching fractions and vibrational energy disposal. In the present study the work is extended to high buffer gas pressure for KrF* and KrCl* at 300 and 230 K with the main objective of measuring the B- and C-state energy separations. Since the ArKrF* and ArKrCl* emission spectra[1b] overlap the KrF(C-A) and KrCl(C-A) bands, experiments were done in N_2 and CF_4 to circumvent trimer formation. The variation of the $I_{ArKrX}/I_{KrX(B,C)}$ ratio with Ar pressure was used to assign the ArKrX* formation rate constants. The three-body quenching of the B and C states seems to be more rapid than that of the D state, leading to an apparent increase in the steady-state KrF(D) and KrCl(D) populations for high pressure of Ar. In addition to interpretation of the high-pressure data for KrF* and KrCl* and discussion of the formation rates of ArKrF* and ArKrCl*, some low-pressure spectra are shown to demonstrate that the general pattern for vibrational relaxation and C → B transfer for KrF* and KrCl* closely resembled the extensively documented XeCl* case.[2] Kvaran, Shaw, and Simons[5] have used the sensitization technique to model the vibrational relaxation of KrF(B). In previous work using the Kr($^3P_1$) and Xe($^3P_1$) sensitization technique, we have identified some mixed rare gas halide trimer spectra,[1b] studied the C-B transfer and vibrational relaxation of XeCl in high v levels,[2] assigned values of the B-C-state energy separation for xenon halides, and made some refinements to the xenon halide C(3/2) and A(3/2) potential curves.[3]

In contrast with the xenon halides, the energy separation between the B and C states of KrX* has not been assigned with confidence. Tellinghuisen and McKeever[6] used their high-pressure (≤750 Torr of Ar) Tesla coil discharge technique to study KrCl*, and they recommended $\Delta E_{BC} = E_C - E_B = 375 \pm 70$ cm^{-1} with

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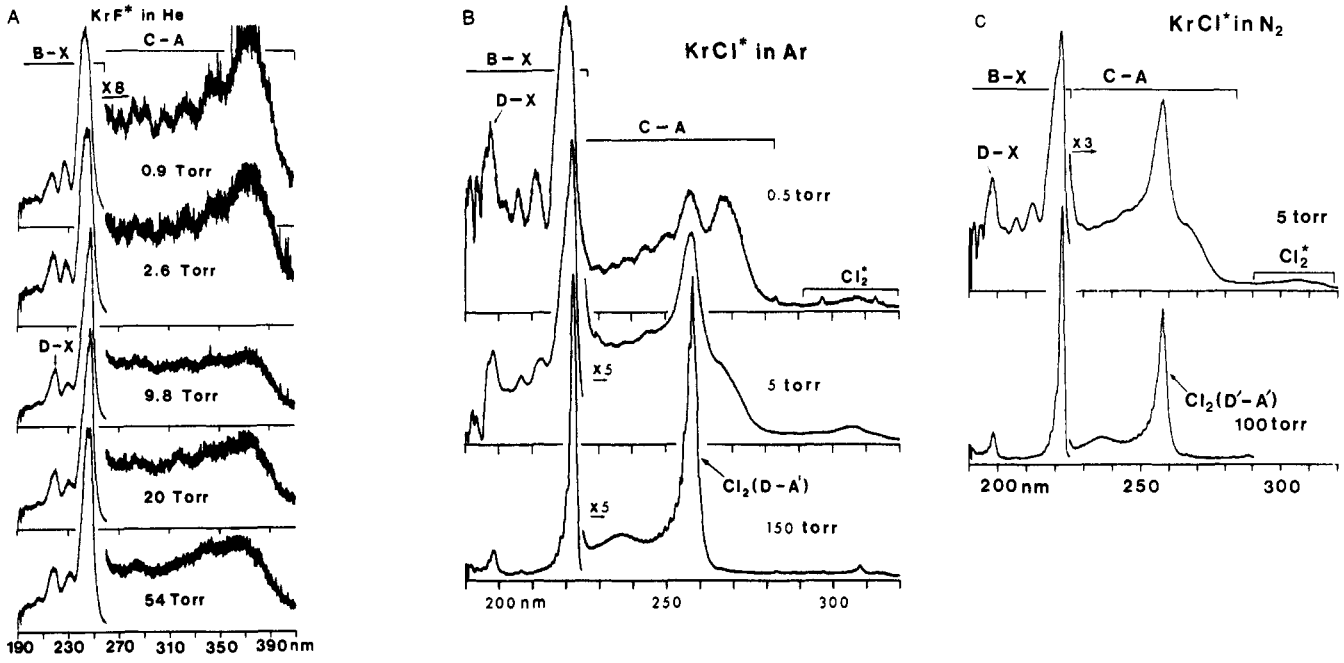


Figure 1. (A) Emission spectra of KrF(B–X) and KrF(C–A) from the $Kr(^3P_1) + F_2$ reaction in low pressure of He buffer gas at 300 K. The KrF(D–X) band, which overlaps the third oscillation of the KrF(B–X) spectrum, is evident in the spectra from experiments above 10 Torr. The KrF(D) contribution is less than 10% relative to KrF(B+C). The KrF(B–X) and KrF(D–X) emission extends into the vacuum ultraviolet. (B) Emission spectra of KrCl(B–X) and KrCl(C–A) from $Kr(^3P_1) + Cl_2$ in low pressure of Ar at 300 K. Note the contribution from $Cl_2(D'–A')$ and other Cl_2^* emission. The low-pressure KrCl(B–X) spectrum extends to shorter wavelength, but this spectrometer was not evacuated and the short-wavelength limit could not be observed. (C) Emission spectra of KrCl(B–X) and KrCl(C–A) from $Kr(^3P_1) + Cl_2$ in N_2 buffer at 300 K. These spectra can be compared to those in (B), which were acquired at similar total pressure, to illustrate the relaxation rates for Ar and N_2.

KrCl(C) being the higher state. LeClavé et al.[7] obtained $\Delta E_{BC}(KrCl) = 380 \pm 40$ cm^{-1} from their value of the coupled radiative lifetime. The ab initio calculated results for KrF*[8] and KrCl*[9] are 806 and 660 cm^{-1}, respectively, with the C state being higher. The calculated ordering[10] for XeF(403 cm^{-1}), XeCl(564 cm^{-1}), XeBr(564 cm^{-1}), and XeI(726 cm^{-1}) are at variance with the currently recommended experimental values: XeF(-775 ± 2 cm^{-1}),[11] XeCl(-90 ± 2 cm^{-1}),[12] XeBr(-110 ± 50 cm^{-1}),[3] and XeI(110 ± 50 cm^{-1}).[3] The first two values are definitive spectroscopic determinations; the last two are based upon C to B intensity ratios. The ΔE_{BC} values for KrX* are expected to be small; the present work based upon the high-pressure intensity ratios supports -120 ± 50 and 0 ± 50 cm^{-1} for KrF* and KrCl*, respectively.

The radiative lifetimes of the B and C states are needed for interpretation of intensity ratios. The calculated radiative lifetimes[8] of KrF* are $\tau_B = 7.0$ ns and $\tau_C = 75.2$ ns. The experimental values for KrF(B), 6.8[13] and 9 ns,[14] are in agreement with the theoretical result. There are no experimental lifetime values for KrF(C). The calculated lifetimes for KrCl(B) and KrCl(C) are 6 and 87 ns, respectively.[9]

The high-pressure KrF(B–X) spectra obtained here have been compared with simulated spectra to improve the potential curves for the KrF(B and X) states and to update the vibrational energy disposal in reactions giving KrF(B). These calculations will be reported separately.[15]

II. Experimental Methods

The resonance lamp and cell were identical with those used for the xenon halide studies[4] at 300 and 230 K, except for replacement of the sapphire window separating the microwave-powered resonance lamp and stainless steel cell (a 2-in. cube) by a MgF_2 window. Teflon O-rings were better than Viton O-rings for interfacing the lamp to the cell because deposition of films on the window separating the discharge and the cell was less serious. The emission spectra were observed by viewing the cell through a quartz window placed on a side perpendicular to the lamp with the same calibrated and computer-interfaced monochromator that was used in the xenon halide studies.[3,4] The spectra to be presented here have been corrected for variation of response with wavelength.

The lamp was filled to $\approx$1 Torr with Kr (Cryogenic Rare Gas Lab; research grade used without further purification). The cold finger on the quartz lamp was cooled with liquid N_2 during operation of the lamp. The microwave discharge, 2450 MHz and $\sim$50–100 W, was adjusted so that the discharge plasma extended to the MgF_2 window. Examination of the output of the Kr lamp with a vacuum monochromator showed no detectable $Kr(^1P_1)$ emission at 116.5 nm. Considering the detection efficiency, there was less than a 5% contribution from the $Kr(^1P_1)$ resonance line (116.5 nm) relative to the $Kr(^3P_1)$ resonance line (123.6 nm). The only other significant emission in the vacuum ultraviolet was the CO(A–X) transition, which caused no difficulty in the present experiments. The more serious problem was N_2 and OH emissions in the ultraviolet, which could be eliminated by evacuation and heating of the lamp and connecting lines. Strong atomic Kr lines ($\lambda \geq 430$ nm) from the microwave-powered lamp precluded observation of product fluorescence in the visible region.

Reagent gases were premixed with Kr and stored in Pyrex reservoirs. After admission of the Kr/reagent mixture, buffer gas was added to the cell to achieve the desired total pressure. Typical concentrations were 30 and 60 mTorr of Kr and reagent, respectively. Pressures were measured with calibrated pressure transducers. It was necessary to pacify the stainless steel cell with F_2 or Cl_2 prior to doing experiments with either reagent. Extreme care must be taken in gas handling when changing reagent gases. For example, after a series of Kr sensitization experiments with Cl_2 were performed, KrCl* emission was still observed for several

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days when experiments with Kr/NF_3 mixtures[16] were attempted.

III. Experimental Results

A. Low- and Intermediate-Pressure Results. The sensitization technique has the advantage, relative to the metastable atom flowing afterglow technique, that experiments can be done at pressures below 0.1 Torr, and collision-free emission spectra can be obtained. The low-pressure spectra consisted predominantly of the B–X and C–A emissions, as assigned in Figure 1. The short-wavelength limits (~150 nm for KrF(B–X) from F_2 and 170 nm for KrCl(B–X) from Cl_2) are obscured by the atmospheric absorption and by the quartz window cutoff in Figure 1. The D–X emission at 220 nm for KrF and 198 nm for KrCl comprises >10% of the B–X and C–A emissions. Precise assignment is difficult because the D–X emission extends below 190 nm; the quoted value is based upon intermediate-pressure data where vibrational relaxation compresses the range of the D–X and B–X emissions. The low-pressure KrF* and KrCl* spectra show the typical oscillatory pattern of the bound–free emission from highly vibrationally excited rare gas halide molecules.[2,5,15,17] In previous work the reactions of $Kr(^3P_1)$ and $Kr(^3P_2)$ with common reagents were qualitatively compared;[1,4] there was little discernible difference between the KrCl* and KrF* emission spectra. The B/C ratios and the vibrational distributions seem to be the same, except for the slightly greater energy (945 cm^{-1}) available from the $Kr(^3P_1)$ reaction. The intensity of the D–X emission is slightly stronger, relative to the B–X emission, for the $^3P_1$ reactions than for $^3P_2$ reactions with X_2.

The excitation-transfer channel does exist for Cl_2, but the total Cl_2* emission is less than 5% of the KrCl* emission. The Cl_2-($D'\,^3\Pi_{2g}$) state is identified from the D′–A′ band at 258 nm in Figure 1B,C; the weak broad band in the 290–320-nm region is probably the transition from the doubled-welled $^1\Sigma_u^+$ state to the repulsive $^1\Sigma_g^+$ state.[18] The structured bands at 287, 295, and 315 nm (see Figure 10) have never been identified. This small branching fraction for Cl_2* formation can be compared to ~25% for Br_2* formation from $Kr(^3P_2)$ + Br_2.[4] The increase in the excitation-transfer channel for Br_2, relative to Cl_2, has been associated with the lower mean vibrational energy $\langle f_V(KrBr)\rangle$ vs $\langle f_V(KrCl)\rangle$.[4] Simons and co-workers[17a,b] have linked the increasing importance of the X_2* product channel with the decline in $\langle f_V(XeX)\rangle$ for $Xe(^3P_2,^3P_1)$ + Cl_2, Br_2, I_2 series, but other factors also may be important.[17c] Predissociation of KrBr* to give Br* does occur, but predissociation of KrCl* requires more vibrational energy than is available from reaction with Cl_2.

It is more difficult to measure the branching fractions for KrF* and KrCl* formation in sensitization than in flowing afterglow experiments. Our attempt to do this by comparing the KrCl* relative emission intensity from Cl_2, SCl_2, $COCl_2$, and HCl for the same Kr concentration gave 1.0:0.18:<0.05:<0.05, which can be compared to the similar trend from $Kr(^3P_2)$ reactions[4] of 1.0:0.38:0.01:<0.01. The same trend in the XeX* product in the branching fractions also exists for $Xe(^3P_2)$ and $Xe(^3P_1)$ atom reactions,[19] and there is little, if any, difference in the dynamics of $Xe(^3P_1)$ vs $Xe(^3P_2)$ or $Kr(^3P_1)$ vs $Kr(^3P_2)$ atom reactions with halogen-containing molecules.

The KrF* and KrCl* spectra in Figure 1 display the characteristic pattern associated with the relaxation of vibrationally excited rare gas halide molecules. As the pressure is increased from 0.2 to 10 Torr, collisions first affect the C state because of the 10-fold longer lifetime, relative to the B state. The KrF(C) molecules that are collisionally transferred to the KrF(B) state lose vibrational energy; thus both the B and C states exhibit vibrational relaxation, which is recognized as a smoothing of the oscillations in both the B–X and C–A transitions. With increase of pressure above 10–20 Torr, the envelope of the B–X emission shifts to longer wavelength, whereas the envelope of the C–A emission shifts to shorter wavelength. The increase in the B-state emission intensity, as the buffer gas is increased to 20 Torr, is shown in Figure 2. The k_B/k_C ratio, obtained from the low-pressure limiting intensity ratio of Figure 2, is approximately 1.3 ± 0.1 for both KrF* and KrCl*. Since the emission below 190 nm could not be included in the integration of the B–X band, the true k_B/k_C ratio will be somewhat larger than the apparent values, 1.3 ± 0.1, and ~1.6 is a good estimate. At low pressure, the

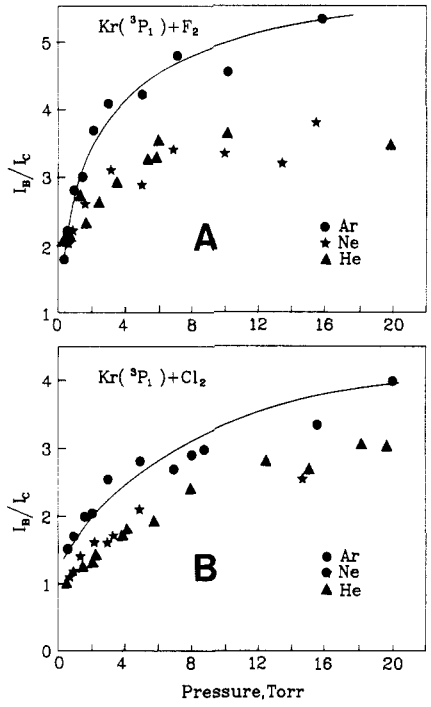


Figure 2. Variation of the B–X to C–A emission intensity ratios from KrF* and KrCl* for low pressures in He, Ne, and Ar. For both cases the C → B transfer, as well as vibrational relaxation, is more efficient in Ar. Since the intensity at λ < 190 nm could not be included in the B–X transition, the low-pressure ratios are lower limits to the true values.

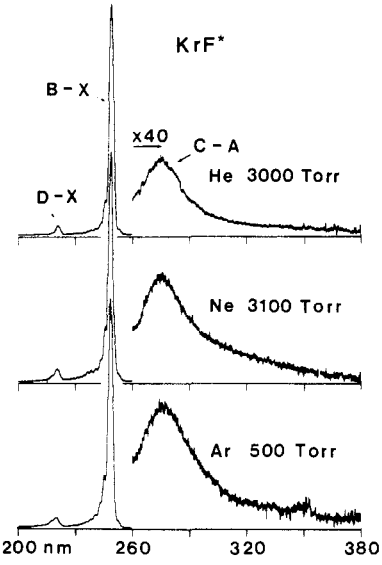


Figure 3. KrF(B–X) and KrF(C–A) emission spectra in high-pressure He, Ne, and Ar at 300 K. The emission from KrF(B,v'=1), which is the shoulder on the blue side of the main band, shows that vibrational relaxation is not complete. The effective vibrational temperature is ~1000 K for each spectrum.

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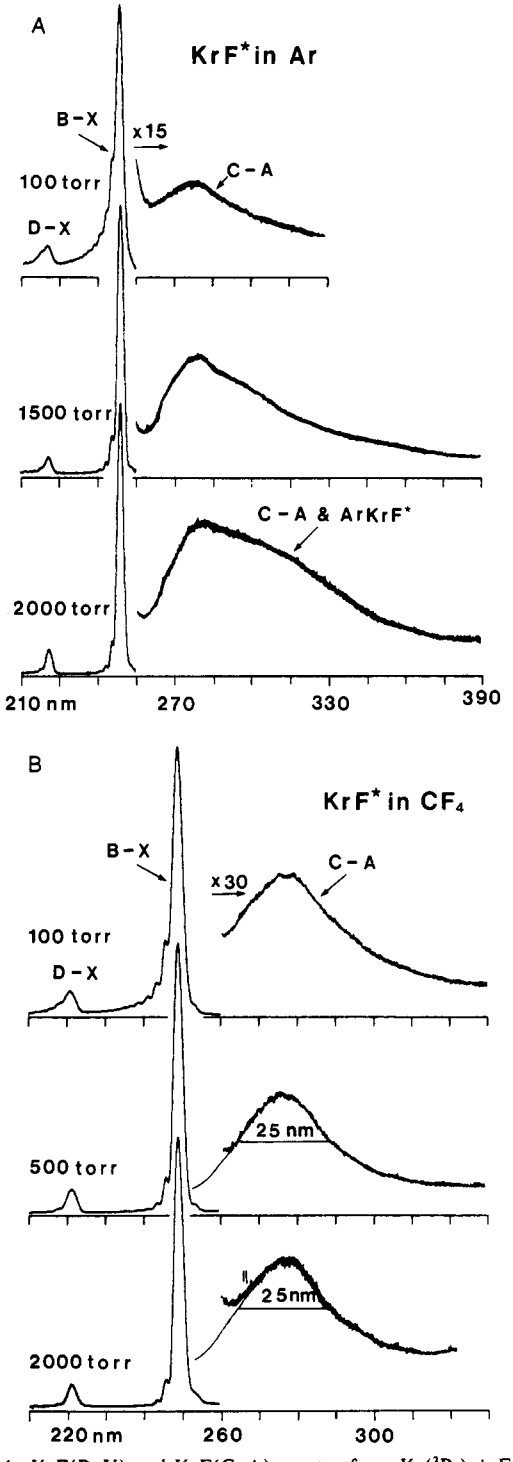


Figure 4. KrF(B–X) and KrF(C–A) spectra from $Kr(^3P_1) + F_2$ in Ar buffer (A) and in CF_4 buffer (B) at 300 K. The spectra in Ar show the development of the ArKrF* trimer emission for the two highest pressures. The spectra from the 500 and 2000 Torr experiments with CF_4 show a 300 K Boltzmann distribution. The KrF(B–X) spectrum in 100 Torr of CF_4 has a vibrational temperature near 1000 K.

kinetics can be treated as transfer from the C state to the B state.[20] A steady-state analysis gives

$$\frac{I_B}{I_C} = \frac{k_B}{k_C} + \left(1 + \frac{k_B}{k_C}\right)\frac{k_{Ar}[Ar]}{A_C} \qquad (1)$$

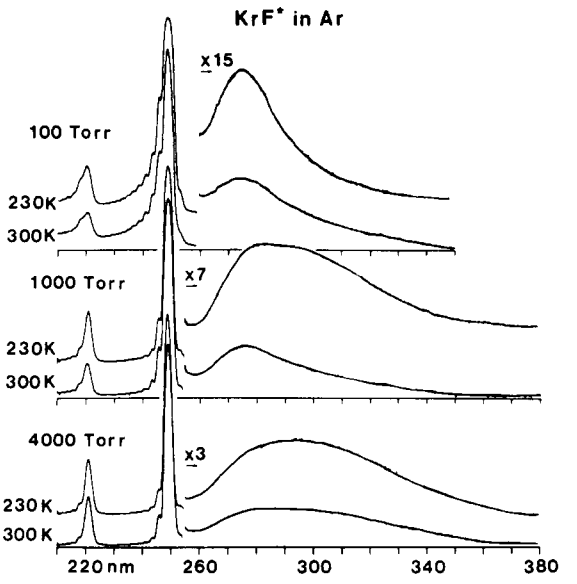


Figure 5. Comparison of KrF* emission spectra in Ar buffer for 300 and 230 K at 100, 1000, and 4000 Torr. The ArKrF* emission is enhanced at the lower temperature.

where k_{Ar} is the transfer rate constant, A_C is the Einstein coefficient, and k_B/k_C is the ratio of rate constants for forming the B and C states. The Ar data gave $k_{Ar} = (2.2 \pm 1) \times 10^{-10}$ cm^3 molecule^{-1} s^{-1} using $\tau_{KrF(C)} = 75.2$ ns. Similar analysis for the KrCl* data gave $k_{Ar} = (6.9 \pm 3) \times 10^{-11}$ cm^3 molecule^{-1} s^{-1} using $\tau_{KrCl(C)} = 87$ ns. Similar rate constants have been found for ArF(C) and XeCl(C) in high v levels.[2,20]

Figure 3 shows some KrF* spectra that have roughly the same vibration distribution ($T_{vib} \approx 1000$ K) for various pressures of He, Ne, and Ar. The vibrational relaxation rates clearly are larger for Ar then for He or Ne. Qualitative modeling[5,15] of the change in the KrF(B–X) spectra vs pressure is in accord with the more quantitative work that has been done with XeCl*.[2] The important points for the present work are (i) the large rate constants for C → B transfer for molecules in high v levels, (ii) the rapid vibrational relaxation for high and intermediate v levels, and (iii) the much slower B → C transfer in low v levels. This pattern taken together with the shorter lifetime of the B state makes it difficult to reach true Boltzmann equilibrium between the B and C states.

*B. High Buffer Gas Pressure Data for KrF**. The emission spectra from $Kr(^3P_1) + F_2$ in high pressures of He, Ne, Ar, and CF_4 are shown in Figures 3–6. The sideband at 244 nm on the KrF(B–X) main peak reflects the $v' = 1$ population and can be used to qualitatively define the vibrational distribution; the width (fwhm) of the C–A band (ca. 25 nm for 300 K conditions) can serve the same purpose for the C state. Boltzmann conditions are better defined with CF_4 than with Ne because of the faster vibrational relaxation and B ⇄ C mixing. The vibrational distribution seems to be nearly Boltzmann at 100 Torr of Ar or CF_4, but the temperature is well above 300 K. At 500 Torr of CF_4 and 1000 Torr of Ar, the vibrational distributions seem to be 300 K. The C–A band narrowed to ~22 nm at 230 K; the same pressures of Ar and CF_4 seem to be required for the vibrational distribution to reach equilibrium at both 230 and 300 K. The vibrational distributions at given pressures of buffer gas are more fully discussed in ref 15.

The growth of the ArKrF* emission at 280–340 nm, which becomes evident for Ar pressure above 1000 Torr in Figures 4A and 5, makes all interpretations about KrF(C) difficult in Ar buffer. The ArKrF* band in Figure 6 was deconvoluted from the KrF(C–A) emission by a point-by-point subtraction of the KrF(C–A) emission from the combined emission, after normalization of each spectrum to the same KrF(B–X) intensity. The deconvoluted ArKrF* spectrum is nearly identical with the one obtained previously.[1b] The ArKrF*/KrF* intensity ratio is shown in Figure 7A as a function of Ar pressure at 300 and 230 K. For the same Ar pressure, the ratio at 230 K is ≈3 times larger than

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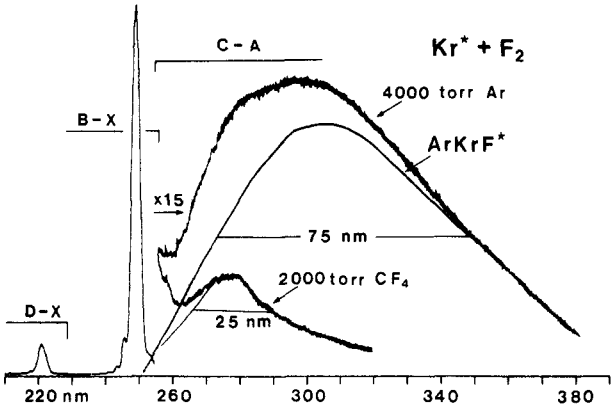


Figure 6. Deconvolution of ArKrF* and KrF(C–A) emission spectra at 300 K using the spectrum from CF_4 to define the KrF(C–A) band.

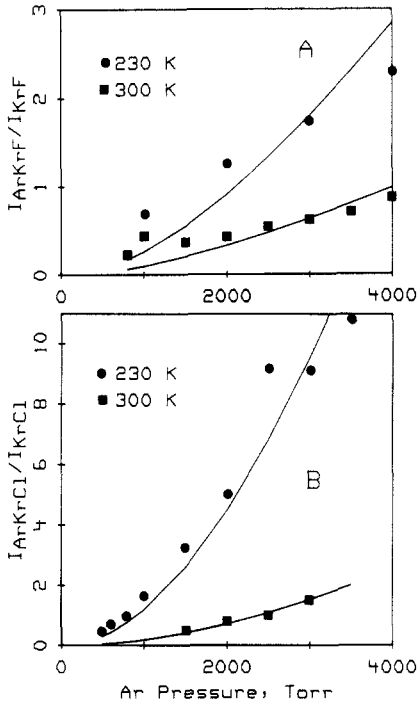


Figure 7. (A) Plots of the ArKrF* trimer emission intensity to the KrF* emission intensity vs Ar pressure at 300 and 230 K. Solid lines are calculated ratios based upon (11). (B) Plots of the ratio of the ArKrCl* trimer emission intensity to the KrCl* emission intensity vs Ar pressure at 300 and 230 K. Solid lines are calculated ratios based upon (11).

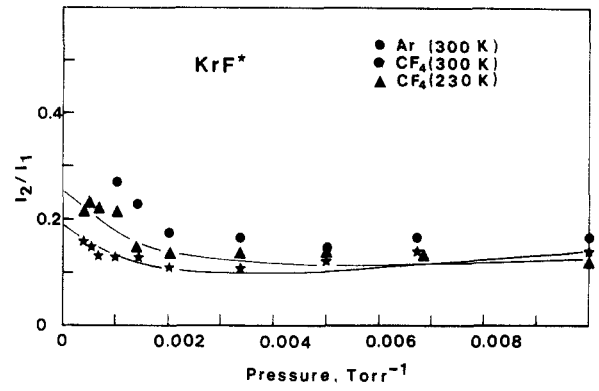


Figure 8. Plots of I_2/I_1 intensity ratio for KrF* (B,C) vs inverse pressure of Ar at 300 K and of CF_4 for 300 and 230 K. The high-pressure data in Ar are less reliable because of the presence of ArKrF* emission and the CF_4 data are preferred; the intercept values are 0.20 ± 0.02 and 0.25 ± 0.02 for 300 and 230 K, respectively.

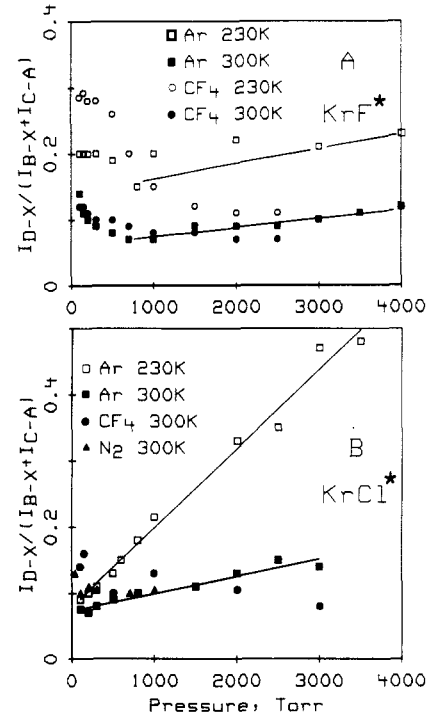


Figure 9. (A) Plots of the ratio of the KrF(D) to KrF(B) + KrF(C) emission intensities vs Ar and CF_4 pressure at 230 and 300 K. (B) Plots of the ratio of the KrCl(D) to KrCl(B) + KrCl(C) emission intensities vs Ar, N_2, and CF_4 pressure at 230 and 300 K.

the ratio at 300 K. The trimer emission dominates the spectrum for pressure above 2000 Torr at 230 K.

One goal was to determine the 300 and 230 K equilibrium values of the KrF(C)/KrF(B) ratio. The ArKrF* emission makes it impossible to obtain accurate intensity ratios from the Ar data, and the most weight must be given to the data with CF_4. The integrated emissions from 225 to 255 nm (mainly B–X emission) and from 255 to 310 nm for I_2 (mainly C–A emission) were labeled I_1 and I_2, respectively. The I_2/I_1 ratio is ~0.1 for a broad range of pressure but then increases slightly for pressure above 500 Torr. The low-temperature CF_4 data show similar behavior, but the intercept is larger. The extrapolated high-pressure intercepts are 0.20 ± 0.02 and 0.25 ± 0.02 for the 300 and 230 K data, respectively. The conversion of these ratios to an ΔE_{BC} value will be delayed to the Discussion section.

An unexpected observation, see Figure 5, was the enhanced KrF(D–X) intensity, relative to KrF(B–X) + KrF(C–A) intensity, with increase of Ar pressure, especially for the low-temperature experiments. As shown in Figure 9A, the KrF(D)/KrF(B+C) ratio first declines as the pressure is increased in both Ar and CF_4 because of two-body quenching of KrF(D); however, for pressure above ≈1000 Torr of Ar the ratio subsequently increases. This increase occurs in the same pressure region that ArKrF* is being formed. Furthermore, the ArKrF* and the KrF(D–X) intensities are both enhanced at 230 K relative to 300 K; see Figure 5. In the Discussion section, we will argue that the apparent increase in the KrF(D–X) emission intensity at high pressure is a consequence of a greater three-body quenching rate of the KrF(B,C) molecules than the KrF(D) molecules. We have no ready explanation for the higher KrF(D) contribution at 230 K in the 50–200 Torr range in *both* Ar and CF_4. This series of experiments was designed to specifically study KrF(D) after most of the other work was completed, and possibly there was more $Kr(^1P_1)$ sensitization than usual, which would favor KrF(D).[20b] The only other explanation would seem to be collisional transfer of KrF(C) to KrF(D) at the lower temperature.

C. High Buffer Gas Pressure Data for KrCl.* The KrCl* spectra from $Kr(^3P_1) + Cl_2$ were recorded in high pressures of Ar and N_2 at 300 and 230 K and in CF_4 at 300 K. Some spectra in Ar and CF_4 are shown in Figure 10. The vibrational relaxation rates are approximately equal for Ar, CF_4, and N_2, judging from the intensity of the sideband at 221 nm. With these buffer gases, the equilibrium vibrational distribution seems established by 500

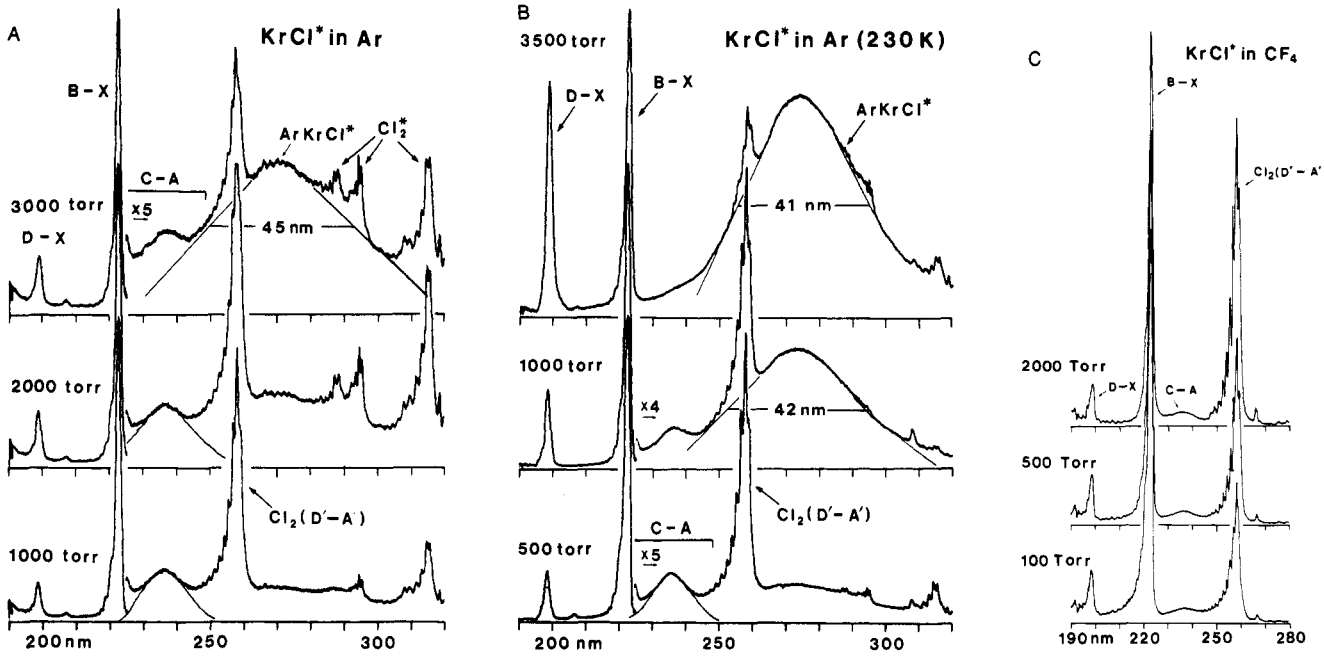


Figure 10. (A) Emission spectra from the $Kr(^3P_1) + Cl_2$ reaction in high-pressure Ar at 300 K. The Cl_2^* emissions were identified in Figure 1B. Note the *increase* in the KrCl(D) intensity with increasing pressure. (B) Emission spectra from the $Kr(^3P_1) + Cl_2$ reaction in high-pressure Ar at 230 K. Note the greatly enhanced ArKrCl* and KrCl(D–X) emission intensities vs experiments at similar pressure for 300 K. (C) KrCl(B–X) and KrCl(C–A) spectra from $Kr(^3P_1) + Cl_2$ in CF_4 at 300 K.

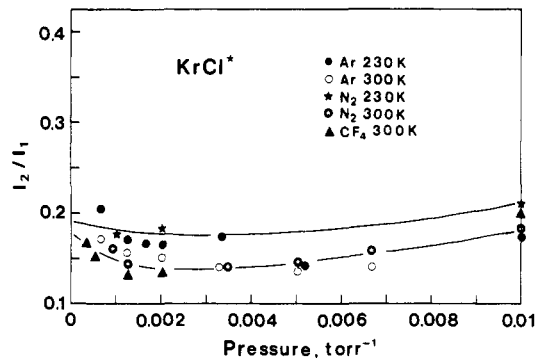


Figure 11. Plot of I_2/I_1 ratio for KrCl* vs inverse pressure of Ar, N_2, and CF_4 at 300 and 230 K. The Ar data are less reliable because of the need to deconvolute from the ArKrCl* emission. The intercept values are 0.18 ± 0.02 at both 300 and 230 K for all gases, except Ar.

Torr at either 300 or 230 K. The fwhm of KrCl(C–A) is ~15 nm at 300 K, and no change was observed at the temperature was lowered to 230 K.

The I_1(210–225 nm) and I_2(225–255 nm) regions were assigned to KrCl(B–X) and KrCl(C–A) emission, respectively. These spectra were planimetered by hand to remove the $Cl_2(D'–A')$ contribution from the KrCl(C–A) band intensity. Determination of the high-pressure limiting I_2/I_1 ratio in Ar is difficult due to the overlapping of the KrCl(C–A) band with both the $Cl_2(D'–A')$ and the ArKrCl* transitions. Since the KrCl* emission was weak in N_2, the CF_4 data should be of the highest quality. The pressure dependence of the I_2/I_1 ratios in different bath gases are shown in Figure 11. The high-pressure intercepts show no discernible temperature dependence, and the intercepts are assigned as 0.18 ± 0.03 (300 K) and 0.18 ± 0.03 (230 K). The effect of the KrCl(B–A) emission on the interpretation of the intensity ratio will be discussed later.

Above 500 Torr of Ar at 230 K or (≈1000 Torr at 300 K), emission from ArKrCl* was observed. Hand planimetering was used to obtain the integrated trimer emission intensity, and the resulting ArKrCl*/KrCl* intensity ratios are shown in Figure 7B as a function of Ar pressure. The intensity ratios are enhanced by a factor of ~6 upon reduction of temperature from 300 to 230 K. An interesting qualitative observation is the larger (by a factor of 3–5) proportion of ArKrCl* than ArKrF* at the same Ar pressure. This must be a consequence of the larger three-body rate constant for formation of ArKrCl*, vide infra.

Figure 9B shows the pressure dependence of KrCl(D–X)/[KrCl(B–X) + KrCl(C–A)] intensity ratio in Ar and N_2. The increase in this ratio as Ar pressure is increased is even more pronounced than for KrF*. This is ascribed to a larger three-body rate constant for the collisionally coupled KrCl(B,C) states than for KrCl(D). Although the N_2 data do not extend to high pressure, there is no obvious enhancement of KrF(D) in N_2 buffer.

IV. Discussion

A. $Kr(^3P_1)$ Sensitized Reactions in the Presence of Buffer Gases. The effect of buffer gas on the total KrX* intensity in the $Kr(^3P_1) + X_2$ system will be discussed first. As common to all sensitization systems, increasing pressure causes Lorentz broadening of the Kr absorption line in the cell relative to the profile of the excitation line.[21,22] This changes the rate of $Kr(^3P_1)$ formation, which makes interpretation of the absolute product intensities difficult unless the buffer gas pressure is the same for different experiments. The resonance $Kr(^3P_1)$ state has a short natural radiative lifetime (3.46 ± 0.09 ns),[23] but the effective lifetime is lengthened by radiation trapping[22] to the microsecond range even for 1 mTorr of Kr, making it possible for quenching to compete with the radiative decay. Upon addition of CF_4 to the sensitized cell, the $Kr(^3P_1)$ state is deactivated by collisions in competition with radiative decay and reactive quenching by Cl_2 or F_2:

$$Kr(^3P_1) \rightarrow Kr + h\nu \quad (2)$$

$$Kr(^3P_1) + F_2 \rightarrow KrF(B,C) + F \quad (3)$$

$$\rightarrow KrF(D) + F \quad (4)$$

$$Kr(^3P_1) + CF_4 \rightarrow Kr(^3P_2) + CF_4 \quad (5)$$

Rate constants for reactions 3–5 have not been measured, so those for the analogous $Kr(^3P_2)$ reactions[24] were assumed: $k_{3+4} = 7.2$

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(22) Mitchell, A. C. G.; Zemansky, M. W. *Resonance Radiation and Excited Atoms*; Cambridge University Press: Cambridge, 1961; pp 106, 192.
(23) Matthias, E.; Rosenbery, R. A.; Poliakoff, E. D.; White, M. G.; Lee, S. T.; Shirley, D. A. *Chem. Phys. Lett.* **1977**, *52*, 239.

$\times 10^{-10}$ cm^3 s^{-1} and $k_5 = 7 \times 10^{-13}$ cm^3 s^{-1}. For typical F_2 pressure (60 mTorr) the rate of loss of Kr($^3P_1$) by reactions 3 and 4 is 1.4×10^6 s^{-1}, and reaction 5 becomes important for $\gtrsim$60 Torr of CF_4. Since reactive quenching of the Kr($^3P_1$) and Kr($^3P_2$) by halogens gives KrX* with comparable efficiency,[1,4,25] reaction 5 actually enhances KrX* formation because the radiative loss of Kr($^3P_1$) is blocked. This mechanism explains why CF_4 is an efficient buffer gas for giving KrF* or KrCl*. If there is a quenching component to (5) in addition to intramultiplet transfer; then the situation is less ideal.

As N_2 was added to the sensitized mixture, the KrX* intensity suffered a greater loss than from addition of CF_4 or Ar. At ~1000 Torr of N_2, the KrCl* intensity was ~5 times weaker than for 1000 Torr of Ar and the C–A emission became too weak to measure. Likely explanations are quenching of KrX*(B,C) and the deactivation of Kr($^3P_1$) by N_2. Since Kr($^3P_1$) is only 945 cm^{-1} above the Kr($^3P_2$) state, quenching by N_2 probably resembles that for the metastable state.

$$Kr(^3P_1) + N_2 \rightarrow Kr(^1S_0) + N_2(B) \quad (6)$$

If the rate constant of Kr($^3P_2$) (0.39×10^{-11} cm^3 s^{-1})[24] is adopted for reaction 6, the deactivation rate will be competitive with the KrF* formation rate at ~12 Torr of N_2. The two-body quenching constant for KrX(B,C) by N_2 has not been measured, but it can be estimated as 3–4 times that for Ar[13] based on analogy to quenching of XeCl* and XeF*.[26,27] Thus, loss of KrF* upon addition of N_2 is mainly from (6) because quenching of KrX(B,C) does not compete with its radiative decay rate, $\tau_{eff}(KrX^*)^{-1}$.

The kinetics of the Kr($^3P_1$) and Kr($^3P_2$) states are more complicated for Ar as a buffer gas than for CF_4 or N_2, since both KrAr* formation and KrF* quenching must be considered. The two- and three-body quenching rate constants for KrF* by Ar have been reported[13] as $(1.8 \pm 0.6) \times 10^{-12}$ cm^3 molecule^{-1} s^{-1} and $(1.1 \pm 0.4) \times 10^{-31}$ cm^6 molecule^{-2} s^{-1}, respectively. Based on these rate constants, ~90% of the KrF* intensity would be lost at 2000 Torr of Ar. The fact that KrF* emission could be easily detected even at 4000 Torr of Ar is not compatible with such a large three-body quenching rate constant. A more plausible estimate of $\sim 8 \times 10^{-33}$ cm^6 s^{-1} can be assigned from three-body formation of ArKrF* at 300 K, vide infra. The three-body rate constant for (8) is the critical step for the loss of KrF* because of the short radiative lifetime of ArKr*.

$$Kr(^3P_1) + Ar \rightarrow Kr(^3P_2) + Ar \quad (7)$$

$$k = 1.3 \times 10^{-13} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1} \quad \text{(ref 28)}$$

$$Kr(^3P_{1,2}) + 2Ar \rightarrow KrAr^* + Ar \quad (8)$$

$$k = (1.0 \pm 0.4) \times 10^{-33} \text{ cm}^6 \text{ molecule}^{-2} \text{ s}^{-1} \quad \text{(ref 29)}$$

$$ArKr^* \rightarrow Ar + Kr + h\nu \quad (9)$$

$$\tau = 333 \text{ ns} \quad \text{(ref 30)}$$

$$ArKr^* + F_2 \rightarrow KrF^* + Ar + F \quad (10a)$$

$$k = 6.0 \times 10^{-10} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1} \quad \text{(ref 30)}$$

$$ArKr^* + F_2 \rightarrow ArKrF^* + F \quad (10b)$$

With 60 mTorr of F_2, the rate of (10a) is 1.2×10^6 s^{-1}, which is only ~40% of that for ArKr* radiative decay. Thus, reaction 8 is a loss mechanism for KrF*. Direct formation of ArKrF* from (10b) has been included in the kinetic model of electron-beam-excited KrF laser[31,32] with an estimated rate constant 3.0×10^{-10} cm^3 molecule^{-1} s^{-1}. However, this reaction has not been isolated and given the magnitude of the excess energy; we favor (10a) as the major product channel. For $k_8 = 1 \times 10^{-33}$ cm^6 s^{-1}, the loss of Kr($^3P_{1,2}$) is not too serious, relative to (3) + (4), for pressures below 1000 Torr of Ar, which is consistent with our qualitative observations. Our data are not consistent with a rate constant as large as 1×10^{-32} cm^6 s^{-1} for (8).[30] According to Young,[28] the three-body rate constant for Kr($^3P_1$) is quite small, $\sim 1.8 \times 10^{-34}$ cm^6 s^{-1}. In Xe($^3P_1$) experiments with added Kr,[21] significant KrXe* emission was only observed for >500 Torr of Kr.

B. ArKrF and ArKrCl* Kinetics.* A mechanism that includes formation, radiative decay, and quenching of diatomic and triatomic rare gas halides was used to explain[2,4] the pressure dependence of I_{KrXeX}/I_{XeX}.[2,4] Steady-state analysis of the mechanism gives

$$\frac{I_{ArKrX}}{I_{KrX}} = \frac{\tau_{ArKrX}^{-1}}{\tau_{eff}(KrX)^{-1}} \left(\frac{k_f[Ar]^2}{\tau_{ArKrX}^{-1} + k_Q[Ar]} \right) \quad (11)$$

where k_f and k_Q are three-body formation and two-body quenching rate constants of ArKrX*, respectively, and $\tau_{eff}(KrX)$ is the effective lifetime of the coupled KrX(B) and KrX(C) states. $\tau_{ArKrF} = 200$ ns and $k_Q = (3–4) \times 10^{-14}$ cm^3 s^{-1} were estimated from $\tau_{Kr_2F^*}$[33] and k_Q of Xe_2Cl^*.[26] Work in the literature,[14] recent two-photon laser-assisted experiments,[34] and our estimate of ΔE_{BC} suggest a coupled lifetime value for KrF*(B,C) of 15 ± 5 ns. By use of these rate constants, $k_f(230 \text{ K}) = (2.0 \pm 0.5) \times 10^{-32}$ and $k_f(300 \text{ K}) = (0.8 \pm 0.3) \times 10^{-32}$ cm^6 molecule^{-2} s^{-1} were estimated for ArKrF* by fitting the experimental intensity ratio to (11); see Figure 7A. The agreement between the experimental and calculated intensity ratios is adequate except at lower pressures where deconvolution of the KrF(C–A) and ArKrF* emissions makes the experimental ratio uncertain.

Similar evaluation of the I_{ArKrCl}/I_{KrCl} ratios with estimated values for $\tau_{eff}(KrCl)$ of 12 ns and $\tau_{ArKrCl} = 200$ ns gives $k_f(230 \text{ K}) = (10 \pm 2) \times 10^{-32}$ and $k_f(300 \text{ K}) = (1.6 \pm 0.3) \times 10^{-32}$ cm^6 molecule^{-2} s^{-1}. The three-body formation rate constants of ArKrF* and ArKrCl* were enhanced by a factor of 2.5 and 6 upon reduction of temperature from 300 to 230 K, which is consistent with theoretical predictions for rare gas halide trimers[35] and three-body recombination reactions in general. Analysis of the I_{KrXeCl}/I_{XeCl} ratios reported previously[1,3] using (11) with $\tau_{eff}(XeCl) = 25$ ns,[36] $\tau_{Xe_2Cl} = 250$ ns,[26] and k_Q[26] = (3–4) $\times 10^{-14}$ cm^3 s^{-1} gives $k_f = 4.6 \times 10^{-32}$ (230 K) and 0.9×10^{-32} (300 K) cm^6 molecule^{-2} s^{-1} for KrXeCl*, which are approximately the same as those for ArKrF*. Our $k_f(300 \text{ K})$ value for ArKrF* formation is a factor of 3 smaller than previous experimental results (2.8×10^{-32} cm^6 molecule^{-2} s^{-1} for $\tau_{eff}(KrF) = 15$ ns[37a] and 4×10^{-32} cm^6 molecule^{-2} s^{-1} [37b]) from an electron-beam experiment. The self-consistency of the rate constants for KrXeCl*, ArKrF*, and ArKrCl* formation, even though some of the lifetimes are not well established, gives strong support for rate constants for mixed trimer formation in the 1.0×10^{-32} cm^6 molecule^{-2} s^{-1} range at 300 K. This is an order of magnitude smaller than the three-body constants for Rg_2X^* (Rg = rare gas) trimer formation from RgX* + 2Rg or RgX* + Rg + Rg'.[33,36] The theoretical result (2×10^{-31} cm^6 molecule^{-2} s^{-1})[38] from a phase space model calculation for mixed

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trimer formation (ArKrF*) is much too large.

C. Pressure and Temperature Dependence of the KrX(D–X)/[KrX(B–X) + KrX(C–A)] Intensity Ratio. The $I_D/(I_B + I_C)$ ratio has a linear dependence on Ar pressure above 1000 Torr; see Figure 9. The removal steps for KrX(B,C) and KrX(D) in bath gas, Q, for a Boltzmann vibrational distribution are listed below; these steps are in competition with radiative decay.

$$KrF(D) + Q \rightarrow KrF(B,C) + Q \quad (12)$$

$$KrF(B,C) + Q \rightarrow \text{quenching} \quad (13)$$

$$KrF(D) + Q \rightarrow \text{quenching} \quad (14)$$

$$KrF(B,C) + 2Q \rightarrow \text{quenching} \quad (15)$$

$$KrF(D) + 2Q \rightarrow \text{quenching} \quad (16)$$

Since KrF(D) is $\approx$5200 cm^{-1} above the KrF(B,C) states, the kinetics are not complicated by transfer from KrF(B,C) back to KrF(D). Steady-state analysis gives

$$\frac{I_D}{I_{B+C}} = \frac{\tau_D^{-1}}{\tau_{eff}(KrF)^{-1}} \times \left(\frac{k_4(\tau_{eff}(KrF)^{-1} + k_{13}N_Q + k_{15}N_Q^2)}{k_3(\tau_D^{-1} + k_{12}N_Q + k_{14}N_Q + k_{16}N_Q^2) + k_4k_{12}N_Q} \right) \quad (17)$$

For pressures below 500 Torr, three-body quenching can be ignored. From studies of XeF[27] and XeCl[36] $k_{12} \approx k_{14} \gg k_{13}$. Thus, the decline of the I_D/I_{B+C} ratio for <500 Torr of Ar and CF_4 can be ascribed to (12) or (14). The increase in I_D/I_{B+C} at >1000 Torr of Ar could be a consequence of the larger three-body quenching rate constant for the KrF(B,C) states than for KrF(D), as Liegel et al. have observed for ArCl* and XeF*.[39] The temperature dependence of I_D/I_{B+C} observed in our study seems to support this argument, since enhanced formation of ArKrF* and ArKrCl* at lower temperature correlates with enhanced I_D/I_{B+C}. The state correlation diagram[33] from RgX(D) and RgX(B,C) with Rg predicts this effect, because only the lowest energy state of the RgX(B,C) pair correlates to the bound trimer $Rg_2X(4^2\Gamma)$. The calculations indicate that the $Rg_2X(7\Gamma)$ state correlating to RgX(D) + X is not strongly bound.[33] Hence, the three-body rate constant would be small.

D. Energy Separations of KrX(B) and KrX(C) States. The I_B/I_C ratio is the equilibrium ratio, rather than a steady-state ratio, if the quenching rates of the B and C states are smaller than the B–C mixing rate in the high-pressure regime, i.e., $k_{BC}[Q] > k_B^Q[Q] + k_B^{2Q}[Q]^2$ and $k_{CB}[Q] > k_C^Q[Q] + k_C^{2Q}[Q]^2$. Generally accepted rate constants[33] for Q = Ar are $k_{BC} \simeq k_{CB} = 10^{-11}$, $k^Q = 10^{-12}$ cm^3 molecule^{-1} s^{-1}, and $k^{2Q} = 10^{-32}$ cm^6 molecule^{-2} s^{-1}. Thus, the two- and three-body quenching rates are less than the B $\leftrightarrow$ C transfer rate even at [Q] = 10^5 Torr. In N_2 and CF_4 the three-body quenching is even less important.

The high-pressure limiting I_C/I_B ratio, I_r, can be used to assign the energy separation in two ways.[3] If the absolute value is used, then (18) applies and a reliable Einstein coefficient ratio (A_C/A_B) is required to deduce ΔE_{BC} from I_r.

$$I_r = I_C/I_B = (A_C/A_B)([C]_{eq}/[B]_{eq}) = A_r \exp(-\Delta E_{BC}/kT) \quad (18)$$

Since the partition function ratio for KrX(B) and KrX(C) is nearly unity, $[C]_{eq}/[B]_{eq} = \exp(-\Delta E_{BC}/kT)$. Alternatively, ΔE_{BC} can be deduced from the intensity ratio at two temperatures.[6]

$$\Delta E_{BC} = kT_1T_2(T_2 - T_1)^{-1} \ln (I_{r_2}/I_{r_1}) \quad (19)$$

The experimental I_2 consists mainly of the C–A transition, but the B–A transition can also contribute to this spectral region. The B–X transition dominates I_1 with negligible C–X transition.[10] The true B/C intensity ratio, I_r, can be obtained by correcting the experimental ratio I_2/I_1 for the B–A transition. The branching ratio, A_{B-A}/A_{B-X}, for KrF* was calculated to be 0.035.[8] Calculation for KrCl* do not exist, but A_r is estimated to be $\sim$0.05, according to the trend in the xenon halide series.[8] Based on this analysis with $I_r(300\text{ K}) = 0.16 \pm 0.02$, $K_{eq}(KrF) = 1.7 \pm 0.2$ and $\Delta E_{BC}(KrF) \approx -110 \pm 20$ cm^{-1}. For KrCl*, $I_r(300\text{ K}) = 0.12 \pm 0.02$, and fortuitously, this ratio also gives $K_{eq} = 1.7 \pm 0.2$ and $\Delta E_{BC}(KrCl) \approx -110 \pm 20$ cm^{-1}. The overlapping of the KrCl(C–A) transition by the $Cl_2(D'–A')$ band causes some uncertainty and the 40% larger A_B/A_C ratio for KrCl* is suspect; it is likely that $\Delta E_{BC}(KrCl)$ is larger than -50 cm^{-1}. With the use of (19), and the intensity ratios at 230 and 300 K, $\Delta E_{BC}(KrF) = -180 \pm 30$ cm^{-1} and $\Delta E_{BC}(KrCl) = 0$ cm^{-1}.

The coupled radiative lifetime of the B and C states provides another measure for K_{eq} that is less dependent on the τ_c^{-1} value.

$$\tau_{eff}^{-1} = (\tau_B^{-1} + K_{eq}\tau_C^{-1})/(1 + K_{eq}) \quad (20)$$

With the calculated lifetimes,[8] $\tau_B = 7.0$ ns and $\tau_C = 75.2$ ns, the effective lifetime of KrF(B,C) is 9.3, 12.8, and 20.7 ns for $\Delta E_{BC} = 200$, 0, and -200 cm^{-1}, respectively. The experimental observations for KrF(B,C) give $\sim 15 \pm 5$ ns[14,34] which seem to support a ΔE_{BC} in the 0 to -150 cm^{-1} range. To our knowledge there are no other measurements for $\Delta E_{BC}(KrF)$; the consensus value from our work seems to be -120 ± 50 cm^{-1}.

The $\Delta E_{BC}(KrCl)$ measured in this study differs significantly from Tellinghuisen and McKeever's[6] value of 375 ± 70 cm^{-1}. This discrepancy is, in part, related to the relatively low buffer gas pressure (<750 Torr of Ar) employed in their experiment. Our results show that the intensity ratio increased slightly for pressures above 1000 Torr, and more than 1000 Torr of Ar is needed to obtain equilibrium B and C concentrations. The I_r value from ref 6 was $\sim$0.09 at 360 K, and extrapolation to the high-pressure limit gave $\sim$0.087. Thus, there is not much actual difference in our experimental ratio at 300 K and theirs at 360 K, see Figure 11, and the difference in ΔE_{BC} arises from the temperature coefficients of I_r. If we use a larger A_{B-A}/A_{B-X}, our K_{eq} would be reduced to some extent. LeCalvé et al.[7] recommended $\Delta E_{BC} = 380 \pm 40$ cm^{-1} from measurement of $\tau_{eff}(KrCl)$. A question in their work is whether the vibrational and electronic state distributions were truly 300 K Boltzmann for pressure below 100 Torr, because the KrCl(B,C) molecules initially were formed in high v levels. Quiñones et al.[36] utilized the two-photon laser-assisted reaction to prepare XeCl(B,C) molecules in Xe and measured the decay constant from 10 to 200 Torr. These measurements give a somewhat larger $\tau_{eff}(XeCl)$ than the data of LeCalvé et al. If there should be a systematic lengthening of lifetimes in the LeCalvé work, then the true $\tau_{eff}(KrCl)$ might be somewhat larger. The KrCl(C) state may be above the B state, but we doubt that the energy difference is as large as 350 cm^{-1}.

Our previous assignments[3] for ΔE_{BC} based upon intensity ratios of the XeX* molecules now can be compared with direct spectroscopic values for XeCl and XeF. Our result for XeF (-610 ± 70 cm^{-1}) is smaller than the exact value (-775 ± 2 cm^{-1}). Substantially higher I_r values ($\sim$5.5 and $\sim$17 at 300 and 230 K, respectively) are required to raise our ΔE_{BC} to -775 cm^{-1}, if the experimental A_r (0.14)[41] is adopted. With benefit of hindsight, inspection of the data suggests that the extrapolations to the high-pressure limit were too conservative; i.e., the B $\rightarrow$ C transfer rate constant is even smaller than we thought. On the other hand, our recommendation for XeCl (-220 ± 40 cm^{-1}) is larger than the exact value (-90 ± 2 cm^{-1}). In this case the experimental I_r value seems reliable; the experimental τ_C probably is too large and the B $\rightarrow$ A branching may be larger than expected.

The inverted energy ordering of the B and C states for the lighter XeX and KrX molecules arises from the stronger configuration interaction between the RgX(B) and RgX(X) states than between the RgX(C) and RgX(A) states.[42] The ΔE_{BC} for KrF* and KrCl* is smaller than for the corresponding xenon

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halides, as would be expected from the weaker configuration interaction for the krypton halides.[42]

V. Conclusions

The $Kr(^3P_1)$ sensitization technique was used to generate KrF* and KrCl* in Kr/F_2 and Kr/Cl_2 mixtures with buffer gases (He, Ne, Ar, N_2, CF_4) at 300 and 230 K. The slow collisional equilibration between the B and C states in He or Ne and the overlapping of the ArKrF* and ArKrCl* emission with the KrF(C-A) and KrCl(C-A) transitions, respectively, in Ar prevented obtaining equilibrium emission intensity ratios from the B and C states in these buffer gases. However, the data in CF_4 and N_2 provide reliable I_{C-A}/I_{B-X} ratios, which support equilibrium constants of ~1.7 and ~1.2 for KrF* and KrCl* at 300 K. Therefore, the long-lived C states must be included in kinetic formulations for these systems. On the basis of the ratio of $I_{ArKrX}/(I_{B-X} + I_{C-X})$ in Ar, three-body ArKrX* formation rate constants for 2Ar + KrX(B,C) were assigned as 0.8 and 1.6×10^{-32} cm^6 s^{-1} for KrF(B,C) and KrCl(B,C), respectively, at 300 K. These values increase by a factor of $\gtrsim 3$ at 230 K. The 300 K three-body rate constant for 2Ar + KrF(B,C) is smaller than the value used in modeling kinetics in the KrF laser.[43] The three-body constant for 2Ar + KrX(D) is substantially smaller than that for 2Ar + KrX(B,C). The secondary processes that affect the KrX* yields when Ar, N_2, and CF_4 are introduced into the sensitization kinetic scheme were discussed.

The equilibrium high-pressure KrF(B-X) spectra from this work have been used in a separate study to improve the KrF(X) and KrF(B) potentials.[15]

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Registry No. KrF, 34160-02-6; KrCl, 56617-29-9; Ar, 7440-37-1; Kr, 7439-90-9; F_2, 7782-41-4; Cl_2, 7782-50-5.

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Kinetics of Partly Diffusion Controlled Reactions. 22. Diffusion Effects on the Kinetics of Excimer Formation[1]

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The kinetics of intermolecular excimer formation are considered in the case where the formation rate is diffusion controlled. As a first step, we assume that an instantaneous monomer-excimer equilibrium occurs: It provides some interesting insights into the influence of the back-reaction on diffusion-controlled reactions. A more realistic model is then developed to provide a complete description of the reaction. Predictions of this model are compared with those of simplified models and with the classic expression for the diffusion-controlled forward rate coefficient.

I. Introduction

Many reactions in liquids have rates sufficiently large that they are affected by the diffusive motion of the reactants. These include aggregation phenomena, enzyme-substrate binding, reassociation of photodissociated pairs, relaxation phenomena following temperature or pH jump, and fluorescence and phosphorescence quenching reactions. When adjacent pairs are assumed to react instantaneously (e.g., colloid aggregation), the reaction is said to be diffusion controlled. When dissociation of adjacent pairs can compete with the rate of reaction, such processes are often referred to as partially diffusion controlled. These diffusion-influenced reactions have been the subject of great interest for more than 70 years.

From a theoretical point of view, it is the early time behavior of these reactions that is of interest. We cite only a small fraction of the papers that have appeared on this topic.[2-15] For a comprehensive review, the reader is referred to the recent monograph by Rice,[16] and some particularly important earlier reviews.[17-19] The short-time behavior of diffusion-controlled reaction has been quite difficult to study experimentally. As a consequence, the number of publications is limited, and there is always a question of the uniqueness of the model to which the data are fit. With the advent of picosecond lasers and improved fluorescence quenching measurement techniques, it is now possible to study these transient effects with greater precision. As a consequence, one has to pay particular attention to the relationship between the reaction one studies and the model one uses to fit the experimental data. For example, the theory is on good footing for

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